

Component Specific Troubleshooting Practical

1. Open your desktop or laptop, locate the RAM slots, and remove and re-insert the RAM module to simulate reseating — then document each step you took and any issues you observed during the process.

Steps:

1. **Shut down the computer** completely and unplug the power cable.
2. **Open the computer case** carefully.
3. **Locate the RAM slots** on the motherboard. They are usually near the CPU.
4. **Release the RAM module** by pressing the locking clips on both sides of the RAM slot.
5. **Remove the RAM** by holding it from the edges. Avoid touching the gold contacts.
6. **Check the RAM and slot** for dust, damage, or any visible issue.
7. **Re-insert the RAM** by aligning the notch on the RAM with the notch in the slot.
8. Press the RAM firmly and evenly until the **locking clips click into place**.
9. **Close the computer case** and reconnect the power cable.
10. **Turn on the computer** and check whether it boots normally.
11. Check the **BIOS/Task Manager** to confirm that the RAM is detected correctly.

Issues observed:

No physical damage was observed. The RAM was reseated properly, and the computer booted normally after the process. If the computer does not boot, the RAM may be loose, faulty, or the slot may have a problem.

2. Create a troubleshooting checklist for diagnosing a faulty GPU in a gaming PC, including at least 5 steps from initial symptom identification to confirming the fix.

Hint: Think about display issues, driver checks, and physical connections.

GPU Troubleshooting Checklist – Gaming PC:-

Step 1: Identify the Problem

First, check the symptom. The PC may show **no display, screen flickering, graphical artifacts, game crashes, or low FPS**.

Step 2: Check the Display Cable

Check the HDMI or DisplayPort cable. Make sure it is properly connected to the **GPU port**. Try another cable or port if available.

Step 3: Check the GPU Connection

Shut down the PC and unplug it. Check whether the GPU is properly seated in the **PCIe slot**. Also check that the GPU power cables are firmly connected.

Step 4: Check GPU Temperature

Use GPU monitoring software to check the temperature. If the GPU is overheating, clean dust from the fans and improve the PC's airflow.

Step 5: Check GPU Drivers

Open Device Manager or the GPU manufacturer's software. Check for driver problems and **update or reinstall the graphics driver** if required.

Step 6: Test the GPU

Run a game or graphics benchmark. Check whether the display is stable and whether the game runs without crashes, flickering, or artifacts.

Step 7: Confirm the Fix

Restart the computer and test it again. If the display works normally, games run properly, and there are **no artifacts or crashes**, the GPU problem is considered fixed.

Result: The GPU was checked through physical connections, display settings, temperature, drivers, and performance testing.

3. Simulate a storage issue by unplugging and replugging a SATA or NVMe drive (SSD/HDD) on a test system, then describe the error messages or BIOS indicators you see before and after reconnecting.

Storage Issue Simulation – SATA/NVMe Drive:-

1. **Shut down** the PC and unplug the power cable.
2. **Locate the SSD/HDD** (SATA or NVMe).
3. **Disconnect** the SATA cable/driver carefully.
4. **Turn on the PC** and open **BIOS/UEFI**.
5. **Observe:** Drive shows **"Not Detected"** or **"No Boot Device Found."**
6. **Shut down** and reconnect/reseat the drive.
7. **Open BIOS again:** The drive should now be detected with its **model and capacity**.
8. **Boot Windows** and confirm the storage is working normally.

4. Given a scenario where a PC randomly restarts, list which component (RAM, GPU, storage) you would check first and explain your reasoning, then outline the specific diagnostic steps you would perform for that component.

1 Check physical seating



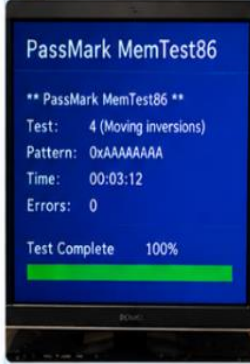
Shut down the PC and unplug the power cable. Open the case and make sure the RAM is properly seated in the slot and the locking clips are closed.

2 Test with one stick



Remove extra sticks and boot with one stick at a time. Check if the restart issue still happens. This helps identify a faulty module or slot.

3 Run MemTest86



Use MemTest86 (bootable USB) to check for memory errors. Let it run for a few passes.

4 Check BIOS / UEFI



Restart and enter BIOS/UEFI to verify that the RAM is detected correctly and running at the right speed.

5 Check temperature & stability



Use a monitoring tool (like HWMonitor) to check temperatures and run the PC for some time. If it restarts again, the issue may be with GPU or storage.